



SCHOOL OF
PROFESSIONAL
STUDIES

2024FA_MSDS_401-0_SEC60

Contact Information

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Response Time: I will respond to emails within 24-48 hours.

Biography:

Irene Tsapara started her teaching career back in 1988, when she taught at University of Illinois while she was finishing her Ph.D. in Computational Learning Theory - (Machine Learning - and Artificial Intelligence). She also holds a Bachelors in Mathematics from the University of Patras and a Masters in Computer Science from the University of Illinois. Before joining Northwestern University, she taught Computer Science, Finance and Advanced Mathematics courses at the University of Illinois, DePaul University and the University of Economics in Athens Greece.

She has extensive experience in the financial industry as a financial engineer for trading systems at Goldman Sachs and Hedge Fund Research where she first implemented the Investable Hedge Fund Indexes. She has also served as a mentor and consultant at different start-ups. She is the founder and president of ZyGen Consulting, a Data Analytics Boutique Consulting Firm, and the Director of a University Ph.D. Program in Data Science.

Her research interests are Learning Algorithms, Data Mining applications in Learning and Financial patterns, and AI and ML applications. Engaged professional with expertise in Artificial Intelligence, Machine Learning, and Neural Networks. Proficient in Python, Keras, and PyTorch, with a keen interest in leveraging these technologies to revolutionize the education sector. Her work is dedicated to preparing the younger generation for the transformative impacts of Artificial General Intelligence (AGI) on both professional and personal spheres, enhancing holistic experiences through AI.

Course Description

This course teaches the fundamentals of statistical analysis. This includes evaluating statistical information, performing data analyses, and interpreting and communicating analytical results. Students will learn to use the R language for statistical analysis, data visualization, and report generation. Topics covered include descriptive statistics, central tendency, exploratory data analysis, probability theory, discrete and continuous distributions, statistical inference, correlation, multiple linear regression, contingency tables, and chi-square tests. Selected contemporary statistical concepts, such as bootstrapping, are introduced to supplement traditional statistical methods. Recommended prior course: MSDS 400-DL Math for Data Scientists.

Course Objectives

At the conclusion of the course, students will be able to:

- Perform statistical analyses.
- Interpret and evaluate statistical information.
- Prepare technical reports.
- Use the language R for data analysis.

Prerequisites

MSDS 400 is a prerequisite for MSDS students who began studies Fall term 2014 and thereafter.

Required Resources

Required Readings

- Black, K. (2013). *Business Statistics*. 8th edition, NJ: John Wiley & Sons, Inc. [ISBN-13: 978-1-118-80048-5] - Available on O'Reilly online for FREE
- Kabacoff, R. I. (2015) *R in Action, 2nd* Shelter Island, NY: Manning Co. [ISBN-13: 978-1617291388] – Available on O'Reilly online for FREE

Optional Readings

- Galvan, J. (2021). *Data Science and Machine Learning with R from A-Z Course*. [Video], O'Reilly

Following is the link for the O'Reilly website. You need to set up an account in this site with your Northwestern email ID and a password.

Link is: <https://learning.oreilly.com/home/>

- Course Reserves from Wilcox and others – Details are in Canvas

MSDS 401 will use the high-level language [R](http://cran.r-project.org/) (<http://cran.r-project.org/>). Instructions are given on the course site. RStudio is not required but is highly recommended for new users of R. Everything students need to do in this course can be accomplished using the standard R console with a plain text editor.

[RStudio http://www.rstudio.com/](http://www.rstudio.com/) provides an integrated environment, and installer packages are easily available online. The installation process is straightforward. Tutorials dealing with R are available on the course site. Students are encouraged to review the *R Tutorial Materials* module on the course site, and to start studying R early in the quarter. There are tests during the quarter dealing with R programming. Check the syllabus and course site for details on date.

Assignment Overview and Grading Breakdown

The students' final grade will be based on the following:

R programming Class Activities	200 possible points	30%
Pre Quizzes	100 possible points	10%
Post Quizzes	100 possible points	10%
2 Data Analysis Projects	200 possible points	20%
• Data Analysis Project 1	100 points	
• Data Analysis Project 2	100 points	
Data Analysis Final Presentation	100 possible points	10%
R Practice	100 possible points	10%
Class Participation	100 possible points	10%

There will be two R programming assignments. Both require free responses, for some questions (ie) not multiple-choice questions. Hence, you need to pay attention and answer them with 'essay-type' responses. No need to include the code in the responses.

Assignment Preparations:

- The Pre Quizzes should be completed by Monday midnight every week.
- The Post Quizzes should be completed by Monday midnight every week.
- The 2 Data Analyses Projects are due on Week 5 and Week 10
- Make sure that you start working on the four assignments 2 weeks before the due dates.

Class Participation:

We will hold a total of 10 Lecture meetings on Tuesdays 11:00 AM to 1:30 PM Central Time. Your attendance and participation is required. Participation in the class discussions, Project presentations, and Group Activities is required and graded. Missing more than two Lectures may result to decrease of your final grade by a whole letter grade.

Grading Scale

Grade	Percentage	Total Points (out of 500)
A	93% - 100%	465 - 500 points
A-	90% - 92%	450 - 464 points
B+	87% - 89%	435 - 449 points
B	83% - 86%	415 - 434 points
B-	80% - 82%	400- 414 points
C+	77% - 79%	385 - 399 points
C	73% - 76%	365 - 384 points
C-	70% - 72%	350 - 364 points
F	0% - 69%	0 - 349 points

Late Work Policy

When stating due dates and times for work, the abbreviation “CST” is used. “CST” means Chicago, IL clock time. This defines “course time”. Canvas will adjust what students see as deadlines according to the time zone specified by the student in personal settings for Canvas. (Thus an 12 pm CST deadline is a 10 pm deadline on the West Coast, and so forth depending on time zone.) Deadlines for all work are stated in this syllabus and posted on the Course Site. This includes exams, reports and participation in the discussions. Assignments are to be submitted prior to the deadlines. **Without prior arrangement, any late assignment will receive a 1% point deduction for each hour late, totaling to a maximum 50% deduction.** For example, ten hours late will result in a 10% point deduction from the total possible assignment points. Prior communication with the instructor is essential.

Do not fall behind. We cover a great deal of material in this course, and falling behind is the primary reason why students have difficulty, particularly toward the end of the course. To that end, the syllabus and course site give due dates for the entire course.

Academic Integrity at Northwestern

Students are required to comply with University regulations regarding academic integrity. If you are in doubt about what constitutes academic dishonesty, speak with your instructor or graduate coordinator before the assignment is due and/or examine the University Web site. Academic dishonesty includes, but is not limited to, cheating on an exam, obtaining an unfair advantage, and plagiarism (e.g., using material from readings without citing or copying another student's paper). Failure to maintain academic integrity will result in a grade sanction, possibly as severe as failing and being required to retake the course, and could lead to a suspension or expulsion from the program. Further penalties may apply. For more information, visit [The Office of the Provost's Academic Integrity page](#).

Instructional Observations

Please note that this course may have a faculty observer for this term. The observer is present exclusively for training purposes, and will not be responsible for, or engage in, any in-class interactions,

student assessment or grading, or any other aspect of course delivery. Please contact the instructor if you have any questions.

Course Schedule

Week 1 – Complete by Sunday

Learning Objectives

After this week, the student will be able to:

- List examples of statistical applications in business.
- Explain the difference between variables, measurement and data.
- Define and compare four different levels of data.
- Construct a frequency distribution and different data displays.
- Construct and interpret two-variable tables and scatter plots.
- Write simple programs using the language R.

Reading

- Black, K. *Business Statistics* (8th edition): Chapter 1 & Chapter 2.
- Course reserves – Chapters 1 & 2 from 10th edition of Business Statistics by Ken Black

R Installation:

- Installation of R and completion of *The Quick Start Guide to R* is expected this week.

Videos:

- Chapter 1 Lecture (Panopto)
- Chapter 2 Lecture (Panopto)
- R Lesson Video

Assignments

- Install R
- Study *The Quick Start Guide to R* and complete the exercises.
- Pre-Quiz by Monday midnight
- Post Quiz by Sunday midnight
- R Practice by Sunday midnight
- Class Activity by Sunday midnight

Week 2 – Complete by Sunday

Learning Objectives

After this week, the student will be able to:

- Calculate and apply measures of central tendency and variability.
- Describe a data distribution using a box-and-whisker plot.
- Interpret graphical data displays.
- Detect outliers using box plots.
- Perform calculations to trim data.

Reading

- Black, K. *Business Statistics* (8th edition): Chapter 3.

Course Reserves:

- Wilcox R. R. *Basic Statistics* Chapter 2: Pages 22-29, Chapter3: Pages 34-45
- Chihara & Hetersberg: *Mathematical Statistics with Resampling and R* Chapter 2 (EDA)

Videos

- Chapter 3 Lecture (Panopto)
- R Lesson Video

Assignments

- Pre-Quiz by Monday midnight
- Post Quiz by Sunday midnight
- R Practice by Sunday midnight
- Class Activity by Sunday midnight

Week 3 -- Complete by Sunday

Learning Objectives

After this week, the student will be able to:

- Describe probability.
- Articulate the different methods of assigning probabilities.
- Understand and apply axioms and properties of probability.
- Compute probabilities under different conditions.
- Understand conditional probability and Bayes' theorem.
- Determine the mean, variance and standard deviation for a discrete variable.
- Solve problems using binomial and Poisson probability distributions.

Reading

- Black, K. *Business Statistics* (8th edition): Chapter 4 & Chapter 5.
- Downey, A. B. *Think Bayes* Chapter 1 pages 1-10 (Course Reserves on the course site.) Here is the link: <http://www.greenteapress.com/thinkbayes/thinkbayes.pdf>

Videos

- Chapter 4 Lecture (Panopto)
- Chapter 5 Lecture (Panopto)
- R Lesson Video

Assignments

- Pre-Quiz by Monday midnight
- Post Quiz by Sunday midnight
- R Practice by Sunday midnight
- Class Activity by Sunday midnight

Week 4 -- Complete by Sunday

Learning Objectives

After this week, the student will be able to:

- Explain what is a probability density function for a continuous variable.
- Compute the expected mean value and variance.
- Describe a standard normal distribution and its properties
- Use the standard normal distribution to find z-scores
- Convert distributions to standard normal.
- Use the normal distribution as an approximation to the binomial distribution.
- Explain different types of sampling plans.
- Explain the central limit theorem.

Reading

- Black, K. *Business Statistics* (8th edition): Chapter 6 & Chapter 7.

Course Reserves:

- Wilcoxon R. R. *Basic Statistics* Chapter 4 Section 7 (Pages 70-76)
- Wilcoxon R.R. *Fundamental of Modern Statistical Methods* – Chapter 3

Videos

- Chapter 6 Lecture (Panopto)
- Chapter 7 Lecture (Panopto)
- R Lesson Video

Assignments

- Pre-Quiz by Monday midnight
- Post Quiz by Sunday midnight
- R Practice by Sunday midnight
- Class Activity by Sunday midnight

Week 5 -- Complete by Sunday

Learning Objectives

After this session, the student will be able to:

- Estimate a population mean and a proportion.
- Define the t-distribution and determine probabilities given degrees of freedom.
- Use the chi-square distribution to estimate a population variance.
- Determine the sample size needed to estimate a population mean and a proportion.
- State what is a confidence interval and how it is used for statistical inference.
- Compute confidence intervals for a mean and a proportion.

Reading

- Black, K. *Business Statistics* (8th edition): Chapter 8.

Course Reserves:

- Wilcox R. R. *Basic Statistics* Chapter 6 Section 5 (Pages 121-129)
- Wilcox R.R. *Fundamental of Modern Statistical Methods* – Chapter 6

Videos

- Chapter 8 Lecture (Panopto)
- R Lesson Video

Assignments

- Pre-Quiz by Monday midnight

- Post Quiz by Sunday midnight
- R Practice by Sunday midnight
- Class Activity by Sunday midnight
- Data Analysis Assignment #1

Week 6 -- Complete by Sunday

Learning Objectives

After this session, the student will be able to:

- Develop one- and two-tailed hypotheses that can be tested.
- Develop test critical regions.
- Reach conclusions based on hypothesis tests
- Explain Type I and Type II errors.
- Perform hypothesis tests on means and proportions.
- Use p-values for hypothesis testing.
- Discuss statistical significance versus practical significance.

Reading

- Black, K. *Business Statistics* (8th edition): Chapter 9 and Chapter 16.

Course Reserves:

- Wilcox R.R. *Fundamental of Modern Statistical Methods* – Chapter 5

Videos

- Chapter 9 Lecture (Panopto)
- Chapter 16 Lecture (Panopto)
- R Lesson Video

Assignments

- Pre-Quiz by Monday midnight
- Post Quiz by Sunday midnight
- R Practice by Sunday midnight
- Class Activity by Sunday midnight

Week 7 -- Complete by Sunday

Learning Objectives

After this session, the student will be able to:

- Develop hypotheses for testing the difference in means or proportions of two populations.
- Use the z-statistic to develop confidence intervals for the difference in two means.
- Perform two-sample t-tests, paired t-tests and construct confidence intervals.
- Develop confidence intervals for the difference in two population proportions.
- Test hypotheses about the difference in variance between two populations.

Reading

- Black, K. *Business Statistics* (8th edition): Chapter 10.

Course Reserves:

- Wilcoxon R. R. *Basic Statistics* pages Chapter 9 Section 9.1 pages 184-193.

Videos

- Chapter 10 Lecture (Panopto)
- R Lesson Video

Assignments

- Pre-Quiz by Monday midnight
- Post Quiz by Sunday midnight
- R Practice by Sunday midnight
- Class Activity by Sunday midnight

Week 8 – Complete by Sunday

Learning Objectives

After this session, the student will be able to:

- Describe what is a designed experiment.
- Use a single factor AOV model for analysis.
- Recognize a randomized block design.
- Explain the advantages of a two-way AOV.
- Compute sums of squares and mean squares
- Use multiple comparison tests.
- Explain what is an interaction.
- Calculate correlations.
- Fit a simple linear regression equation.

Reading

- Black, K. *Business Statistics* (8th edition): Chapter 11 & Chapter 12 Sections 12.1-12.3.

Course Reserves:

- Wilcox R. R. *Basic Statistics* Chapter 10 Section 10.1 pages 210-217.

Videos

- Chapter 11 Lecture (Panopto)
- Chapter 12 Lecture (Panopto)
- R Lesson Video

Assignments

- Pre-Quiz by Monday midnight
- Post Quiz by Sunday midnight
- R Practice by Sunday midnight
- Class Activity by Sunday midnight

Week 9 – Complete by Sunday

Learning Objectives

After this session, the student will be able to:

- Explain a simple linear regression model.
- Determine the equation of a simple linear regression line.
- Specify the two parameters of a straight line.
- Discuss the risks of extrapolation.
- Perform inference about regression coefficients.
- Calculate the Pearson product-moment correlation coefficient.
- Calculate standard errors and confidence intervals for regression coefficients.
- Test the overall model.
- Assess Model Adequacy.

Reading

- Black, K. *Business Statistics* (8th edition): Chapter 12 Sections 12.4-12.7 & Chapter 13.

Course Reserves:

- Wilcox R. R. *Basic Statistics* Chapter 8 Section 8.3 pages 172-176.

Videos

- Chapter 12 Lecture (Panopto)
- Chapter 13 Lecture (Panopto)
- R Lesson Video

Assignments

- Pre-Quiz by Monday midnight
- Post Quiz by Sunday midnight
- R Practice by Sunday midnight
- Class Activity by Sunday midnight

Week 10 – Complete by Sunday

Learning Objectives

- No new learning objectives.

Assignments

- Presentations of Data Analysis 2 Project

There is a practice test with solutions posted in Canvas under Week 10. ***Practice Problems*** are posted in the weekly modules. These are practice problems and carry no point value. They may be attempted at any time during the course.